HireMind Deep Learning Research Pack

Five papers and a connected experiment plan

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1 One platform and one research component

Senior develops HireMind, the recruitment platform. DL studies a small model that receives CV text and a job description and predicts No Fit, Potential Fit or Good Fit. After evaluation, its output could become one evidence source in Senior's scoring workflow. Practical challenges, explanations, storage and recruiter decisions remain separate platform functions.

The current task is CV–job fit classification, not extracting individual skill names. Fit labels support the former; skill extraction would require different annotations and experiments.

The rule we will follow

Reuse relevant Senior work; adapt it when necessary; create what is missing. Label each item by its benefit to Senior, DL or both. Separate completed work from planned features and measured results from hypotheses.

Item	Decision	Benefit and status
Senior motivation and Gan review	Reuse with corrections	Both • Reviewed; no HireMind performance established
Earlier BERT reference	Expand into architecture analysis	Both • Reviewed; model training remains planned
Turc, Hinton and Q8BERT papers	Add missing technical evidence	Both • Reviewed; methods not yet implemented
Senior workflow and interface	Retain platform context	Senior; future DL demonstration • Integration planned
Dataset audit and controlled experiments	Create new research work	DL directly; potential Senior benefit • Audit completed; training pending
Other Senior papers	Keep outside this five-paper DL set	Senior • Earlier verification limitations still apply

What is complete and what remains

Complete: five paper reviews, dataset audit, idea comparison and experiment plan. Hams reports instructor approval of the dataset and two-person team, and creation and sharing of GitHub and course-template Overleaf projects. Technical Report is selected. Pending: pilot, training, measurements, editor access, calendar confirmation, student revision and final new-chat review.

These materials are preparation support. The course limits AI-generated content to 25% of submitted work and requires disclosure. Rawan and Hams must verify the evidence and write the submission in their own words; this pack does not satisfy that limit merely by including a disclosure.

2 Reading notes for the five selected papers

Each note answers the assignment's paper questions. Results below belong to the cited papers, not to HireMind. Feasibility judgments and next questions are our proposed adaptations.

Paper 1 Gan and colleagues 2024

Question	Critical reading
Problem and method	Automate resume screening using LLM sentence classification, grading, summarization and agents.
Contribution and data	Integrated screening study; 1,000 predominantly IT resumes with 78,668 classified sentences; separate grading labels largely generated by GPT-4.
Baseline and result	Table 1: instruction-formatted LLaMA2-7B-chat reaches 87.73 F1, versus 86.67 for BERT Large.
Limitations	Different tasks from our classifier. Table 10's speed comparison uses estimated human reading time. The grading data description inconsistently gives 838 total versus 500 training plus 383 test resumes.
Colab feasibility	Full reproduction is unsuitable for our budget; use the recruitment context, not its large-model training setup.
Our next question	Can a compact classifier supply useful CV–job fit predictions at lower deployment cost?
Reuse and evidence	Adapt Senior summary • Both. Sections 4–6; Tables 1 and 10; PDF pages 8–16.

Paper 2 Devlin and colleagues 2019

Question	Critical reading
Problem and method	Learn transferable bidirectional text representations using a Transformer, masked language modeling and next-sentence prediction.
Contribution and data	One pretrained architecture can be fine-tuned for multiple tasks. Pretraining uses BooksCorpus and Wikipedia; evaluation includes GLUE and SQuAD.
Baseline and result	Table 1: BERT Base scores 84.6% on MNLI matched, compared with 82.1% for OpenAI GPT.
Limitations	General NLP results do not establish recruitment validity. Large pretraining cost and a 512-token input limit matter for long CV –job pairs.
Colab feasibility	Fine-tune an existing checkpoint; do not reproduce pretraining.
Our next question	How much task quality can a compact BERT retain under a fixed input budget?
Reuse and evidence	Expand earlier Senior reference • Both. Sections 3–5, Table 1 and Appendix A; PDF pages 3–9 and 13–14.

3 Compact models and distillation

Paper 3 Turc and colleagues 2019

Question	Critical reading
Problem and method	Build effective compact language models through pretraining followed by task-specific teacher distillation.
Contribution and data	Compares 24 model sizes and examines transfer-data size and domain. Tasks include GLUE and Amazon review classification.
Baseline and result	Table 3: a 6-layer student's six-task development meta-score rises from 82.8 with pretraining plus fine-tuning to 84.4 with pretrained distillation.
Limitations	That result is not for BERT-Tiny or recruitment. Benefits depend on task and transfer data; reported hardware speedups cannot be copied to our CPU.
Colab feasibility	Use a released compact checkpoint and one teacher; omit large-scale pretraining and the 24-model sweep.
Our next question	Does distillation improve the same small classifier on limited CV–job pairs?
Reuse and evidence	New detailed review • Both. Sections 3–6; Tables 1, 3 and 4; PDF pages 3–9.

Paper 4 Hinton and colleagues 2015

Question	Critical reading
Problem and method	Transfer knowledge from an expensive model or ensemble into a deployable student using softened class probabilities.
Contribution and data	Temperature-based distillation, optionally combined with hard-label loss. Experiments cover MNIST, speech and specialist image models.
Baseline and result	On MNIST, the smaller hard-label model makes 146 errors; its distilled version makes 74; the larger teacher makes 67.
Limitations	Not a BERT or recruitment study. Some exploratory missing-class experiments adjust biases using the test set; we must not copy that evaluation practice.
Colab feasibility	A small teacher–student adaptation is feasible to investigate; reproducing the large speech or image experiments is outside scope.
Our next question	Do teacher probabilities improve generalization beyond the same student trained on hard labels?
Reuse and evidence	New • Both. Sections 2–4 and 8; PDF pages 2–5 and 8.

4 Quantization and the combined research gap

Paper 5 Zafrir and colleagues 2019

Question	Critical reading
Problem and method	Reduce BERT deployment cost with 8-bit quantization-aware training, quantizing embedding and fully connected operations while retaining selected higher-precision operations.
Contribution and data	Evaluates quantization on GLUE tasks and SQuAD; compares FP32, quantization-aware training and dynamic quantization.
Baseline and result	MRPC F1: 90.00 FP32, 89.56 QAT and 87.88 dynamic quantization. Results are development-set averages over five runs. The paper reports about fourfold weight compression.
Limitations	Hardware acceleration is conditional. Its QAT procedure differs from our post-training method; its fourfold result is not a measured HireMind result.
Colab feasibility	Investigate CPU dynamic Linear quantization plus weight-only embeddings. This is post-training quantization; QAT remains outside the current experiment.
Our next question	Can the quantized student retain the distillation gain and reduce serialized model size?
Reuse and evidence	New • Both. Sections 2–4, Table 1; PDF pages 2–4.

What the five papers support together

The recruitment paper motivates the application. BERT supplies the architecture foundation. Turc and Hinton motivate teacher-guided training. Q8BERT supports investigating precision reduction and highlights potential quality loss. Together they justify an experiment; they do not guarantee a positive result or establish that recruitment distillation is a new invention.

Our proposed research question

Can teacher-guided distillation improve a pretrained compact CV–job fit classifier, and can post-training quantization reduce its storage cost while retaining that improvement?

What changes in Senior if the experiment succeeds

A future backend endpoint could accept a CV–job text pair and return three class probabilities, a predicted class and a model version. Senior could display this result alongside its other evidence. It must not be presented as verified skills, behavioral assessment, a calibrated hiring probability or a final hiring decision. API-based functions can remain elsewhere in the platform.

Two corrections to the earlier plan

Quantization reduces precision, not necessarily parameter count, token count or theoretical operation count. Distillation can add training cost even if the deployed student is inexpensive. Report those costs separately instead of claiming unmeasured energy savings.

5 Three candidate ideas and our selection

All three ideas use CV–job fit labels, a pretrained BERT-Tiny baseline, job-grouped validation and the same final test policy. Their numerical targets are exploratory, not reported results.

Idea	M0 then M1 then M2	Targets and main difficulty
A Teacher-guided training Selected	M0: hard-label fine-tuning. M1: same student, adding teacher distillation. M2: INT8 Linear layers and 8-bit embedding weights in M1.	H1: +1 macro-F1 point. H2: at least 60% smaller, retaining at least half the positive gain. Difficulty: teacher quality and extra training.
B Class-weighted training Backup	M0: ordinary cross-entropy. M1: inverse-frequency weighted cross-entropy using training counts only. M2: quantized M1.	H1: +1 macro-F1 point. H2: at least 60% smaller, retaining half the gain. Difficulty: weighting may hurt other classes; its direct literature support would need expanding.
C Longer text context Alternative	M0: maximum 128 tokens. M1: same architecture trained with 256 tokens. M2: quantized M1 at 256 tokens.	H1: +1 macro-F1 point. H2: at least 60% smaller than M1, retaining half the gain. Difficulty: longer input increases runtime and changes the input budget.

Why select A

A most directly uses the selected papers and isolates a training change while preserving the student architecture and input length. It could produce a small deployable component for Senior. B is cheaper if the teacher is ineffective, but changing to it would require a revised hypothesis and supporting method reference. C is simple but risks making the model slower before compression.

Exact model roles

Role	Definition
Supporting teacher	google-bert/bert-base-uncased with a three-class head, fine-tuned only on the training portion. Select its checkpoint using validation; cache training logits.
M0	google/bert_uncased_L-2_H-128_A-2 with a three-class head. Train on hard labels. Approximately 4.4 million parameters; measure the actual task model.
M1	Same pretrained student initialization and head seed as M0. Train using 0.5 × hard-label cross-entropy + 0.5 × T^2 × KL(teacher probabilities || student probabilities), with both softened at T.
M2	Quantize each M1 CPU copy: dynamic INT8 Linear modules plus 8-bit weight-only Embedding modules. Use a dedicated embedding configuration. Layer normalization and unsupported operations remain FP32. No additional training.

M1 does not receive extra hard-label epochs by continuing from trained M0. M0 and M1 start from matched initializations and have equal update budgets. The teacher is additional support, not a replacement for the required M0 baseline.

6 Dataset audit and evaluation boundary

Instructor-approved dataset: cnamuangtoun/resume-job-description-fit. Inspected revision: 08978e21714984bb417547d2c0f9b477f5298163. File-structure checks were performed on 13 October 2026; they are not model evaluation.

Check	Observed result
Schema	resume_text, job_description_text, label
Original split	6,241 training rows; 1,759 test rows; 8,000 total
Labels	No Fit: 4,000; Potential Fit: 2,000; Good Fit: 2,000
Unique raw text	643 CVs and 351 job descriptions across both files
Repeated pairs	7,993 unique raw-text pairs; seven excess pair occurrences
Conflicting labels	Six identical raw-text pairs carry conflicting labels, affecting twelve rows
Split overlap	476 identical CV texts occur in both files; no identical job descriptions overlap
Empty fields	No empty or whitespace-only fields in the three required columns
Documentation	Repository contains the two CSVs and .gitattributes; no dataset card or licence found in this revision. Label origin and validation remain unverified.

Cleaning and splitting plan

Quarantine the twelve conflicting rows and collapse the remaining duplicate pair, yielding 7,987 unique non-conflicting pairs. Preserve the official test jobs. Reserve approximately 20% of original training-job groups for validation using seed 42. Keep each job entirely within one split.

Exact and whitespace-normalized checks are complete. Inspect near duplicates before freezing the split. Save hashes, indices and label counts. Do not choose the split using model performance. If validation lacks a class, use a documented, predetermined grouped stratification procedure.

What our test can and cannot show

This evaluates new job descriptions with potentially familiar CVs. It does not measure generalization to wholly unseen candidates, real hiring outcomes or fairness across demographic groups. Related CV–job pairs are not independent observations; report job-grouped uncertainty where possible.

Dataset approval and limitations

The instructor approved this dataset for the project, as reported by Hams. Original provenance, licence and label origin remain undocumented. Audit a training-only sample using a written fit rubric; do not assume expert labels. Keep raw CV text out of the repository. Approval does not establish real-world hiring validity.

The audit script reproduces data checks without printing CV contents. No training pilot has been run. The next check covers checkpoint loading, embedding quantization, save/reload and prediction validity on Colab.

7 Experiment and hypotheses

Fair training and model selection

Use the same three labels, tokenizer, maximum length of 256 tokens, longest-first pair truncation and preprocessing for teacher, M0 and M1. Report the fraction truncated and inspect which sections are lost. Use AdamW, effective batch size 16, weight decay 0.01 and a maximum of three epochs, selecting checkpoints by validation macro-F1.

Proposed equal search budget: M0 learning rates {2e-5, 5e-5, 1e-4, 3e-4}; M1 learning rates {5e-5, 1e-4} × temperatures {2, 4}, with the hard/soft loss weight fixed at 0.5. Search at seed 42, then rerun the chosen M0/M1 settings at seeds 43 and 44. Use the same update cap. Teacher search uses learning rates {2e-5, 5e-5}. Log every trial and all teacher costs.

Numerical hypotheses

Let F0, F1 and F2 be mean final-test macro-F1 over paired seeds, expressed on a 0–100 scale, and let S1 and S2 be serialized model sizes measured consistently.

H1: $F1 - F0 \geq 1.0$ percentage point. H2: $S2 \leq 0.40 \times S1$, $F1 - F2 \leq 1.0$ point, and $F2 - F0 \geq 0.5 \times (F1 - F0)$, with $F1 > F0$. Targets are exploratory. If M1 does not improve on M0, the claim of retaining an improvement fails; report compression separately.

The revised 60% storage-reduction target includes 8-bit embedding weights and dynamic INT8 Linear modules. Embeddings require their own weight-only configuration. Remaining FP32 values and quantization metadata prevent assuming exactly fourfold compression. Latency and memory are measured outcomes, not promised gains.

Measure	Protocol
Quality	Macro-F1 primary; accuracy, per-class precision/recall/F1 and confusion matrix secondary. Freeze choices before final test evaluation.
Size and parameters	Use the same serialization approach, excluding optimizer states. Report bytes and MiB separately. Record logical parameter count and quantized module coverage.
CPU latency	Same CPU, thread count, batch size 1, input lengths and software. Warm up 30 calls; time 200 repeated calls. Report median and p95, plus a separate tokenization-inclusive measurement.
Memory	Fresh process for each model; report peak resident memory, baseline process memory, and peak GPU training memory where available.
MACs and FLOPs	Use the same input shape and counting convention; disclose unsupported operations. Quantization can leave operation counts unchanged while reducing bit width.
Uncertainty and cost	Report paired-seed mean and standard deviation. If computing intervals, bootstrap job groups. Record teacher training, student training and logit-cache generation time separately.

Pilot before committing to full runs

On Colab, load the checkpoints and check quantization, save/reload and finite three-class predictions. Run a training-only subset and validate on a separate subset. Record time, memory, macro-F1, truncation and CPU latency. Pin compatible versions. Free Colab is the starting point; resource availability is not guaranteed.

8 Work plan and submission checks

Course week	Deliverable	Proposed lead
10	Document approved dataset limitations; audit labels; freeze grouped split; confirm shared workspace access.	Rawan: audit; Hams: setup; both: approval
11	Baseline pilot; teacher and M0 runs; record feasibility.	Hams: training; Rawan: data checks
12	M1 distillation and matched comparisons.	Hams: model; Rawan: error analysis
13	M2 quantization, CPU measurement and repeatability checks.	Rawan: measurement; Hams: export
14	Freeze settings, evaluate final test, interpret failures and limitations.	Both
15	Student-written report, slides, poster and optional Senior integration demo.	Both; each documents actual contributions

The instructor approved the two-person team; Technical Report is selected. The table proposes course Weeks 10–15, ending in Week 15. This allocation needs confirmation against the actual teaching calendar; it is not a verified current-week or deadline claim.

Risk and fallback decisions

Weak teacher: check validation quality and report negative transfer. Unsupported embedding quantization: use Linear-only quantization as a documented fallback and revise H2 before test evaluation. Quality loss: report it honestly. Inspect truncation using training/validation only. Preserve the dataset documentation limitations despite instructor approval.

Proposal checklist

Requirement	Current position
Five papers and research question	Reading notes and question prepared.
Three ideas, M0/M1/M2, H1/H2	Prepared; numeric targets explicitly exploratory.
Dataset, fair comparison, feasibility	Audit and plan prepared; instructor approval reported. Documentation limitations remain; pilot not run.
1–2 page proposal plus references	Revised ACL-style draft supplied. Hams reports using the approved Overleaf template; compile revised text there to confirm layout.
Roles, timeline, risks and AI statement	Drafted; roles are proposals, not claims of completed student contributions.
New-chat AI review and appendix	Not performed. Use the exact Blackboard reviewer prompt in a fresh chat after student revision; attach the resulting review and fix major issues.
Online links and submission	Hams reports GitHub/Overleaf creation and sharing completed. Editor access pending. No submission, final results or deployment claimed.

GitHub and Overleaf have been created and shared according to Hams. The final reviewer appendix is still required after student revision. Confirm editor access and course weeks. The final Technical Report also requires named contributions, presentation, poster and a portfolio URL in the submission README.

9 Sources and reading record

[1] Chengguang Gan, Qinghao Zhang and Tatsunori Mori. 2024. Application of LLM Agents in Recruitment: A Novel Framework for Resume Screening. arXiv:2401.08315v2. https://arxiv.org/abs/2401.08315v2

[2] Jacob Devlin, Ming-Wei Chang, Kenton Lee and Kristina Toutanova. 2019. BERT: Pre-training of Deep Bidirectional Transformers for Language Understanding. NAACL-HLT, pp. 4171–4186. https://aclanthology.org/N19-1423/

[3] Iulia Turc, Ming-Wei Chang, Kenton Lee and Kristina Toutanova. 2019. Well-Read Students Learn Better: On the Importance of Pre-training Compact Models. arXiv:1908.08962v2. https://arxiv.org/abs/1908.08962v2

[4] Geoffrey Hinton, Oriol Vinyals and Jeff Dean. 2015. Distilling the Knowledge in a Neural Network. arXiv:1503.02531v1. https://arxiv.org/abs/1503.02531

[5] Ofir Zafrir, Guy Boudoukh, Peter Izsak and Moshe Wasserblat. 2019. Q8BERT: Quantized 8Bit BERT. arXiv:1910.06188v2; EMC2 workshop. DOI: 10.1109/EMC2-NIPS53020.2019.00016. https://arxiv.org/abs/1910.06188v2

Implementation and data resources

Dataset: https://huggingface.co/datasets/cnamuangtoun/resume-job-description-fit
Student checkpoint: https://huggingface.co/google/bert_uncased_L-2_H-128_A-2
Teacher checkpoint: https://huggingface.co/google-bert/bert-base-uncased
Quantization API: https://docs.pytorch.org/docs/main/generated/torch.ao.quantization.quantize_dynamic.html
ACL style source: https://github.com/acl-org/acl-style-files
Course template link: https://www.overleaf.com/read/mknhysghvcgh

These are software, data and formatting resources, not additional research papers in the selected five. Pin library versions and checkpoint revisions after a successful pilot.

Reading and verification scope

The complete extracted text of the five downloaded papers was reviewed, including their references and appendices where present. Key result tables were also inspected visually. The notes identify the exact versions and relevant pages. Reading does not independently reproduce the authors' experiments or validate all their claims.

Gan: 18 PDF pages. BERT: 16 PDF pages, including appendices. Turc: 13 PDF pages. Hinton: 9 PDF pages. Q8BERT: 5 PDF pages. Evidence and dataset checks were performed on 13 October 2026. The original Senior review remains unchanged; this pack replaces the earlier ten-entry list for the current five-paper DL plan.